



Electricity from Carbon emissions

Scientists from Korea have developed a water-based fuel cell that can convert CO₂ into electrical energy and hydrogen. <https://digit.in/feb19-51>



Oceans are warming faster

According to researchers oceans are absorbing heat 40% faster compared to the previous estimates. <https://digit.in/feb19-52>

THE DRIFTING MAGNETIC NORTH

The magnetic north pole, towards which all compass needles on the Earth point, is shifting its position and this shift is accelerating.

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he Earth actually has a number of north poles, and all of them are not at a fixed point. The Earth

rotates towards the east, leading to periods of days and nights. The points at which the axis of rotation intersects with the surface of the Earth is called the geographic north pole and the geographic south pole. At the south pole, there is a land mass, while the north pole has drifting ice sheets. The exact location of the geographic poles, also known as the terrestrial poles, is not constant. It wobbles by a few meters, following some regular patterns, as well as an irregular pattern. The geographic poles are also referred to as the “true” poles.

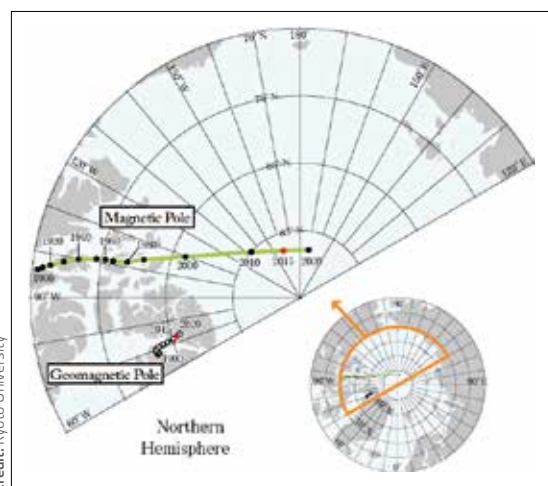
THE MAGNETIC POLES

Apart from the geographic poles, there are the magnetic north and south poles. This is where the compass needles point towards at any given point in time. At the magnetic north pole, the

compass points straight down. When early compasses were invented, the end that pointed to the north was marked with an N. Eventually, scientists figured out that the needles were attracted to the opposite poles, but the convention stuck. So compass needles point to the magnetic north pole, which is actually the south pole of the magnetic field of the Earth. Imagine a line between the

magnetic north pole and the magnetic south pole. This line is at an angle of about eleven degrees to the axis around which the Earth rotates, or a line between the geographic north pole and the geographic south pole. Now, the line between the magnetic north pole and the magnetic south pole, is not a line that passes through the center of the Earth. This means that the magnetic

north and south poles are not located exactly at opposite sides of the Earth. The distance between the magnetic north pole and the geographic north pole is about 500 kilometres. The distance between the magnetic south pole and the geographic south pole, is much more, at 2,858 kilometres. If you were actually standing at the geographic north or south pole, your compass would still point towards the magnetic poles.



Credit: Kyoto University

The drift of the magnetic and geomagnetic north pole